MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP- ZOOLOGY (HONOURS) V SEMESTER PRACTICAL EXAMINATION 2026

MODEL QUESTION PAPER (2024 Admission onwards)

MG5DSCZGY301- CELL BIOLOGY AND MOLECULAR BIOLOGY

Duration: 2 hrs. **Maximum Marks: 35**

Certified Record – 10 Marks

Part A [Skill] [CO-5]

I. Squash preparation of onion root tip (10 marks)

Preparation and identification of any two mitotic stages (3+3=6 marks)
Comment on (1+1=2 marks)
Sketch and label the parts (1+1=2 marks)

OR

Prepare human blood smear (10 marks)

Preparation and identification of any two leucocytes (3+3=6 marks)
Comment on (1+1=2 marks)
Sketch and label the parts (1+1=2 marks)

II. Prepare a smear of human buccal epithelium and identify the Barr body and Comment on (2+2=4 marks)

Part B [Understand] [CO-5]

III. Write down the procedure involved in DNA isolation (3marks)

Part C [Understand] [CO-5]

IV. Identify and comment on:(Any 2- Cell organelles/meiotic stages using slides or diagrams; Identification- ½mark; comment -1½ marks) (2x2=4marks)

V. Identify and write notes on:(Any 2- Molecular composition / structural orientation / functional significance of DNA, DNA replication, RNA different types, Polytene chromosomes using models or diagrams; Identification- ½mark; comment -1½ marks)

(2x2=4marks)

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